Available online at www.sciencedirect.com

ScienceDirect

Procedia Computer Science 233 (2024) 960–969

Procedia Computer Science

www.elsevier.com/locate/procedia

5th International Conference on Innovative Data Communication Technologies and Application (ICIDCA 2024)

Towards Detection of Network Anomalies using Machine Learning Algorithms on the NSL-KDD Benchmark Datasets

Amol D. Vibhute[a,*], Chandrashekhar H. Patil[b], Arjun V. Mane[c], Karbhari V. Kale[d]

[a]*Symbiosis Institute of Computer Studies and Research (SICSR), Symbiosis International (Deemed University), Pune-411016, MH, India.*
[b]*School of Computer Science, Dr. Vishwanath Karad, MIT-World Peace University, Pune, MH, India.*
[c]*Department of Digital and Cyber Forensic, Government Institute of Forensic Science, Nagpur, MH, India.*
[d]*Professor and Vice-Chancellor of Dr. Babasaheb Ambedkar Technological University, Lonere-402103, MH, India.*

amolvibhute2011@gmail.com

Abstract

In the present era, everyone is connected via the Internet for sharing digital information. The digital data is stored using the cloud technology. However, cloud technology is speedily increasing the volume of digital information and network intrusions. In this case, safeguarding the cloud data is essential for several purposes. Therefore, the present study emphasizes developing the network intrusion detection system using the benchmark NSL-KDD datasets. The ensemble learning-enabled random forest algorithm was proposed and implemented to select the most suitable features. The network intrusion detection and classification have been done using three machine learning models: support vector machine (SVM), logistic regression, and K-nearest neighbour's (KNN) with 87.58, 88.86, and 98.24% validation accuracies. Thus, the present study approach can be used in real-time cyber-attack detection and monitoring.

© 2024 The Authors. Published by Elsevier B.V.
This is an open access article under the CC BY-NC-ND license (https://creativecommons.org/licenses/by-nc-nd/4.0)
Peer-review under responsibility of the scientific committee of the 5th International Conference on Innovative Data Communication Technologies and Application

Keywords: Random forest model; Cyber-attacks; Network anomaly detection; AUC-ROC curves, One-hot encoding method; K-nearest neighbours.

1. Introduction

Information technology proliferates with increased reliance on computer systems in today's interconnected world. This leads to an unprecedented growth in the volume of digital information with high dimensional characteristics [1]. Cloud computing technology has recently emerged to store and share digital information remotely in a more flexible, scalable, and inexpensive way for individuals and businesses. However, the use of the cloud is rapidly increasing, and multiple users are generating high-dimensional data [2]. Alongside this, the high-volume cloud data can rise in cyber threats, attacks, and network traffic that exploit vulnerabilities in cloud systems.

1877-0509 © 2024 The Authors. Published by Elsevier B.V.
This is an open access article under the CC BY-NC-ND license (https://creativecommons.org/licenses/by-nc-nd/4.0)
Peer-review under responsibility of the scientific committee of the 5th International Conference on Innovative Data Communication Technologies and Application
10.1016/j.procs.2024.03.285

Moreover, cloud data processing is also challenging with high-dimensional data. As digital information grows, the prerequisite for proactive and sophisticated approaches to safeguard information and infrastructure becomes more persistent. Therefore, cloud security is needed to ensure sensitive data and detect and protect against evolving cyber threats. In this case, developing effective models for detecting network intrusions and mitigating cyber threats and cyber-attacks is vital to secure network data. Furthermore, creating an effective intrusion detection system that detects zero-day attacks is necessary. Therefore, network anomaly detection models can detect and mitigate zero-day or unknown attacks [3]. However, the intrusion detection systems are majorly classified into three categories: (1) signature-based, (2) anomaly-based, and (3) hybrid-based. In signature-based systems, the intrusions can detect attacks based on particular patterns (signatures) available in the existing database.

On the other hand, anomaly-based systems find network abnormalities to detect threats or intrusions. The hybrid-based approach combines the signature and anomaly systems to find the attacks [4]. However, network anomaly detection systems are majorly used in detecting abnormal activities in computer networks or cloud computing. The anomaly-based systems can be developed using statistical, knowledge-based, or artificial intelligence-based methods, especially machine learning [5]. However, the statistical and knowledge-based systems do not accurately detect network anomalies and zero-day attacks [5]. Therefore, machine learning-enabled models have previously been implemented to detect and classify potential network attacks with benchmark datasets automatically.

For this reason, the cybersecurity research community has developed several near-real-time benchmark datasets to develop and test the machine learning models that detect network anomalies. For instance, the NSL-KDD dataset [6, 7] is one of the standard datasets set to build a robust intrusion detection system that can discern regular network traffic from suspicious or malicious activities by applying machine learning models. In the NSL-KDD datasets, denial of service (DoS), probing, unauthorized access like R2L, and U2R attacks are present and utilized to prepare the machine learning models [6]. In this context, using machine learning models on the NSL-KDD datasets holds great promise to enhance computer network security significantly [7]. Moreover, machine learning-enabled intrusion detection systems successfully detect cyber-attacks and mitigate the ever-evolving cyber threats in the complex cloud environment and interconnected information technology globe [8, 9]. The current article aims to create a novel network anomaly detection system using the machine learning models implemented and tested on the benchmark NSL-KDD datasets. The significant contributions of the present paper are:

1. Proposes an innovative framework to detect and classify the network anomalies with the highest detection accuracy rate.
2. Explores the random forest-based feature selection model in which we develop several decision trees and choose the essential features with primary voting (averaging).
3. Develop machine learning models selecting the best features to enhance the intrusion detection performance.
4. Evaluate the proposed machine learning models on NSL-KDD datasets with a confusion matrix, the area under the receiver operating characteristic curves (AUC-ROC), and several performance measurements such as prediction accuracy, recall, precision, and f1-scores.
5. Compare the results obtained via the present study with earlier standard literature to show the effectiveness of the proposed approach.

The rest of the article is separated as follows. Section two reviews the recent standard literature and finds the research limitations in a prior study. The proposed framework and methodology are adopted in section three. This section also focuses on the datasets used, their pre-processing, and machine learning models with evaluation measurements. The results obtained via the proposed approach are discussed in section four. This section also compares the present study's outcome with recent standard literature and shows its importance. The last section concludes the current research and insights into the future scenario.

2. Literature review

Previous studies have identified network intrusions with NSL-KDD datasets using machine learning methods. For instance, the study [10] applied an enhanced J48 algorithm with other machine learning methods on NSL-KDD training and testing datasets. This study [10] achieved only 90.02% accuracy on test datasets and 76.23% on the

entire train and test datasets using the enhanced J48 algorithm. The other machine-learning models have resulted in very limited NSL-KDD datasets [10]. The research done in [8] has proposed the machine learning-enabled intrusion detection system on NSL-KDD datasets and compared it with the generative adversarial network. The implemented models such as KNN (91.33%), SVM (90.38%), decision tree (92.23%), random forest (91.28%), and artificial neural networks (90.93%), have obtained limited detection rates with restricted datasets [8]. In addition, the authors [1] implemented random forest, artificial neural networks, KNN, and SVM algorithms on NSL-KDD datasets with only 80.21, 83.45, 82.98, and 83.35% accuracies without using the feature selection approach. However, the accuracy of models was enhanced up to 96% after using the selection approach [1].

On the other hand, in the study [5], a signature and anomaly-fused intrusion detection system has been proposed and implemented on NSL-KDD datasets using a one-class SVM method with 83.24% accuracy. However, SVM and KNN models have produced only 69.52 and 79.4% accuracy in their studies [5]. Similarly, the system was proposed for detecting the host-based intrusions using the 2-class SVM and bagging algorithms. The models were executed on the NSL-KDD datasets and obtained only 69.52 and 97.51% accuracies. In addition, some machine learning methods were also implemented with limited precision [11].

Some studies have also focused on deep learning models for detecting and classifying network anomalies. For instance, the study [12] proposed a deep learning-based BiLSTM model and machine learning-based KNN method to detect network intrusions with 95.4 and 78.1% accuracy, respectively. In this case, they used KDD CUP 99 datasets, which is more straightforward than NSL-KDD [12]. Moreover, the study uses and runs several deep learning models on NSL-KDD datasets and achieves a maximum of 88.13% testing accuracy [4].

It is observed from the literature review that the accuracy still needs to be improved on NSL-KDD datasets for detecting network intrusions and classifying accordingly. In addition, no intrusion detection system provides 100% detection rates due to daily new attacks. The earlier approaches used traditional feature selection methods that could not benefit the machine learning models. Moreover, the proposed techniques in previous studies either take more training times, produce a limited detection rate, select inappropriate features or generate more false recognition rates. Therefore, one standard system is needed to detect recent anomalies/attacks with a high success rate or reduce the false recognition rate with limited features and training timings. Thus, we covered these limitations in the present study by proposing a machine learning-enabled feature selection approach with the highest accuracy. In this case, we used three machine learning models to identify and classify network anomalies implemented on NSL-KDD datasets with high detection rates.

3. Materials and methods

3.1. Datasets

The present study uses the standard benchmark NSL-KDD datasets designed at the University of New Brunswick in Canada by the Canadian Institute for Cybersecurity (CIC) to evaluate the intrusion detection systems [13]. This dataset is the updated version of the KDD Cup 1999 dataset to avoid redundant and duplicate records. Thus, the classification models will not be biased. There are four significant intrusions found in NSL-KDD datasets, such as DoS, R2L, U2R, and probe, besides ‘normal’ network traffic. The NSL-KDD-Train dataset used in the present research is divided into 75% (125,973 records) for the training and 25% (22,544 records) for validation purposes. The overfitting of the models was avoided in the validation process during the training step [4]. In addition, the NSL-KDD-Test+ dataset was used to test the validated models. Forty-one attributes were present in the NSL-KDD dataset in three categories; the rest were in numeric format labelled with ‘normal’ traffic or intrusions.

3.2. Pre-processing

The pre-processing is vital to normalize the data for further processing and modelling machine learning algorithms to obtain the highest recognition rate. In the raw datasets, there were some missing, null or not number (NaN), inconsistent values with irrelevant and redundant columns that do not provide any information. Therefore, we first identified these distortions and eliminated them from the datasets to clean the data. In addition, three categorical features were present in the original datasets that were converted into the numerical format using the one-hot encoding method

[14, 15]. Subsequently, the dataset is normalized to confirm that the numerical features were normalized (standardized) to bring them to a standard format and enhance the performance of the models, reducing the error rate during the training phase. The min-max normalization method [15] is used to standardize the data implemented using Eq. (1). Thus, the data is normalized between 0 and 1.

$$A_{Norm} = \frac{A - A_{min}}{A_{max} - A_{min}} \tag{1}$$

Where, A_{Norm} and A are the normalized data and normalizing the data from the data set, respectively.

3.3. Feature selection with random forest

Feature selection is an essential phase in intrusion detection to enhance the accuracy of the machine learning models. The feature selection algorithms provide the best possible features to avoid redundant dataset information. In the current research, we employed the random forest [16] model due to its popularity in selecting the appropriate features and classification tasks. The random forest uses an ensemble learning approach to merge the numerous decision trees and classify the best features [16]. Moreover, the random forest reduces the dimensionality of the imbalanced datasets and measures the rank of the components via the information gain. This information can be employed in selecting the best features [16]. We chose the scikit-learn library to run the random forest method in determining the most valuable features. Firstly, the datasets were divided into training (70%) and testing (30%) with 300 n_estimators (decision trees) and 100 random states. Then, the best nine features were selected from the random forest model, setting 0.05 constant threshold values. In this case, the random forest model has taken almost 349 seconds to train the model.

3.4. Proposed machine learning models

3.4.1. Support vector machine (SVM)

The SVM is a statistically supervised machine learning algorithm used in the present study due to its high performance in detecting several types of intrusions in cybersecurity. The SVM finds a hyperplane to simplify binary classification using the high-dimension space. The primary target of SVM is to select the ideal linear optimal hyperplane that can best separate the maximum margin between two classes. It uses some portion of the data to train the system and picks numerous support vectors that signify the training data [1, 3, 5]. Subsequently, these support vectors can be used in developing the SVM models. Mathematically, the SVM algorithm is implemented using Eq. (2) of a hyperplane to classify the two classes.

$$f(x) = (w, x) + b \tag{2}$$

Where, w and b describe the weights and bias, respectively. In this case, x is denoted as a regular class if f(x) is positive or else intrusions (see Eq. (3)).

$$f(x) = \begin{cases} +1, \; if \; x \geq 0 \; (Normal \; class) \\ -1, \; if \; x \leq 0 \; (Intrusion) \end{cases} \tag{3}$$

Here, the x on or above the hyperplane is considered *normal* class otherwise *intrusions*. The SVM offers various kernels to detect intrusions. The present study has chosen the linear kernel due to its simplicity and performance in binary classifications with fewer training samples.

3.4.2. Logistic regression

Logistic regression is a statistical and supervised machine learning method for binary classification. It works on the relationship between a dependent binary variable and one or multiple independent numerical variables. The logistic regression model uses the logistic (sigmoid) function (Eq. (4)) to convert the linear combination of input features into 0 and 1 values [17].

$$z = \frac{1}{1 + e^{-z}} \tag{4}$$

Where z=the linear combination of the input features implemented using Eq. (5).

$$Z = \beta_0 + \beta_1 x_1 + \beta 2 x_2 + \ldots \beta_n x_n \tag{5}$$

Where $\beta_1, \beta_2, \ldots \beta_n$ are the coefficients allied with each feature $x_1, x_2, .., x_n$.

The logistic regression determines the maximum likelihood estimation to select the optimal values for the coefficients β that best fit the training data. Our model chooses uniform weights related to all classes, 100 iterations, and 0.5 threshold values to train the model efficiently and classify data points into one of the two classes.

3.4.3. K-nearest neighbour's (KNN)

The KNN is also one of the supervised non-parametric machine learning algorithms used in intrusion detection due to its simplicity and high performance. It makes predictions by choosing the appropriate k-nearest data values in the training datasets. In this case, KNN uses the Euclidian distance method to measure the distance between data points. Then, it calculates the distance between the new data point and all data points in the training dataset and selects the k-nearest neighbours based on these distances. The distance between two data attributes can be computed using Eq. (6). However, the KNN method does not use the training step but works during predictions [1, 8, 17]. In our case, the k-neighbors' values were set to five with 30 iterations. The sklearn library implements the KNN algorithm in the present study.

$$Ecludian\ distance = \sum_{i=1}^{n} \sqrt{ai - bi} \tag{6}$$

Where a and b are the two data points in Euclidian distance space.

3.5. Evaluation Measures

The offered machine learning models were assessed using methods such as accuracy, precision, recall, and f1-score, which were calculated via Equations (7), (8), (9), and (10), respectively.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{7}$$

$$Precision = \frac{TP}{TP + FP} \tag{8}$$

$$Recall = \frac{TP}{TP + FN} \tag{9}$$

$$F1 - S\,ccore = 2 * \frac{Recall * Precision}{Recall + Precision} \tag{10}$$

Where TP is truly positive, predict one, and the actual is also one. TN is true negative predicts zero, and natural is also zero. An FP is false positive indicates one, but the essential is zero, and an FN is false negative predicts zero but is one consistently [2, 11].

4. Results and discussion

The proposed algorithms were executed on personal Windows 11 Home version 22H2 operating systems, 16 GB RAM sizes, and the AMD Ryzen 5-5500U processor with Radeon 2.10 GHz graphics card. The Python 3.8 programming language in Anaconda Jupyter Notebook has been used to implement and test the machine learning models. The Python libraries such as *numpy*, *sklearn*, *matplotlib*, *pandas*, and *seaborn* were used in the Python run of the models. The raw downloaded NSL-KDD data in *.csv* format was pre-processed in the initial step. The missing NaN and inconsistent values were corrected with redundant columns.

Moreover, we cleaned the raw data and converted them into a suitable format to perform the machine learning approaches. The frequency count of every attack type is then calculated to select the best features. In the NSL-KDD dataset, a total of four intrusions were present. However, detecting these intrusions from the forty-one features is a

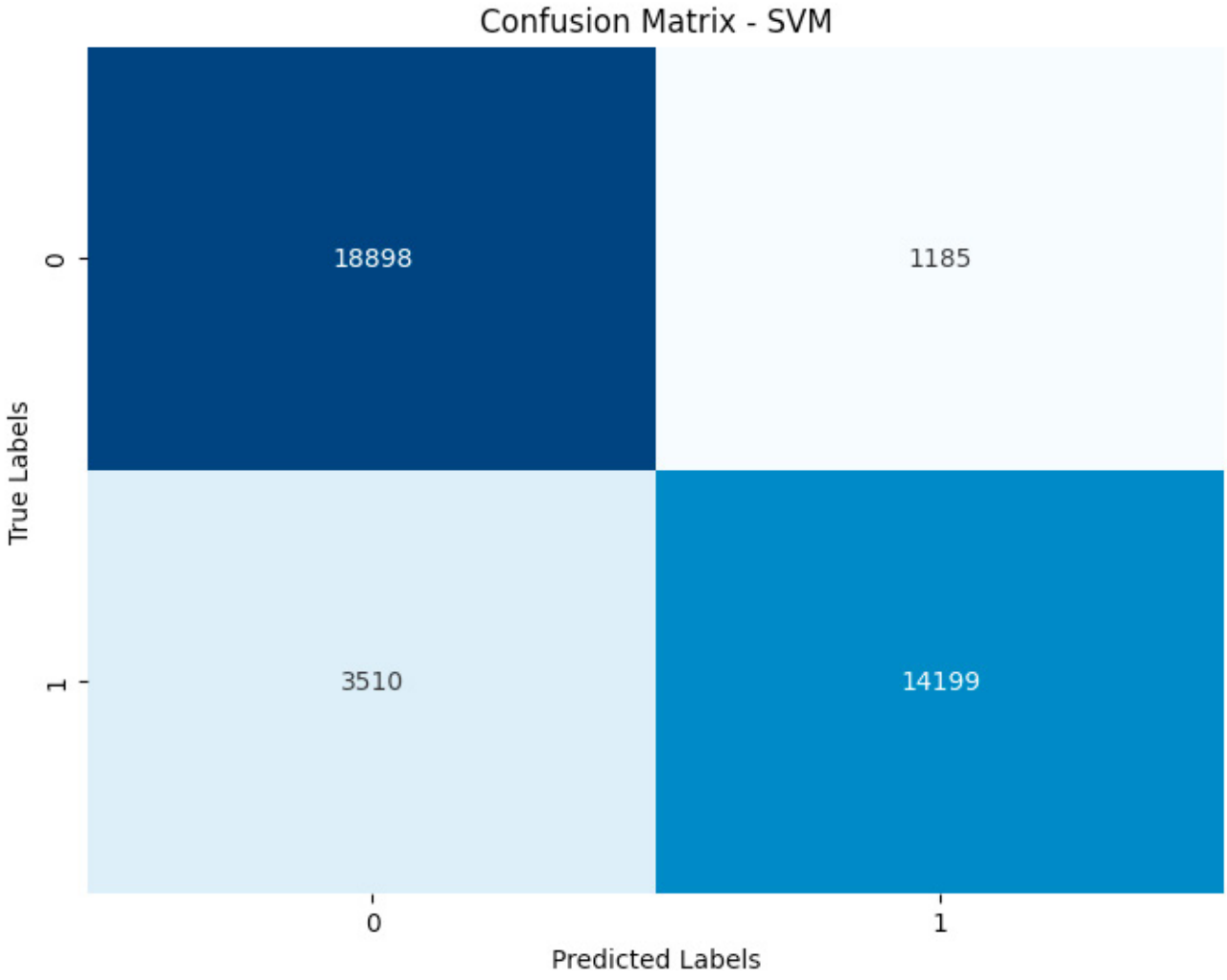


Fig. 1. The confusion matrix generated using the SVM method executed on the validation dataset (where, 'normal': 0, 'anomaly': 1).

tedious task. Hence, we applied an ensemble learning-enabled random forest algorithm [16] and chose only the nine best features without losing the vital information. As a result, we obtained only important features from forty-one and trained the classification models. In selecting the best features via the random forest model, we set a 0.05 threshold value and obtained 99.54% accuracy with selected features.

Subsequently, we propose and execute the three machine learning models, such as SVM, logistic regression, and KNN, as discussed in section 3.4. The normalized data was first separated into training (75%) and validation (25%) phases, with 125,973 and 22,544 records, respectively. The proposed SVM model has been trained with a linear kernel, forty-two random_states, ten penalty parameters, and a gamma value. The SVM model has provided 87.58% prediction accuracy. In the case of logistic regression and KNN models, we achieved 88.86% and 98.24% accuracies, respectively. The resultant validation accuracy and their misclassifications are demonstrated in Table 1 for all implemented methods.

The proposed machine learning algorithms have been assessed using the confusion matrix, 10-fold cross-validation and performance measurements such as accuracy, precision, recall, and the f1-score with the AUC-ROC curve. The outcomes acquired are demonstrated in Table 2 separately for each method applied in detecting the intrusions. The applied classification models outperformed the best features selected via the random forest algorithm from the NSL-

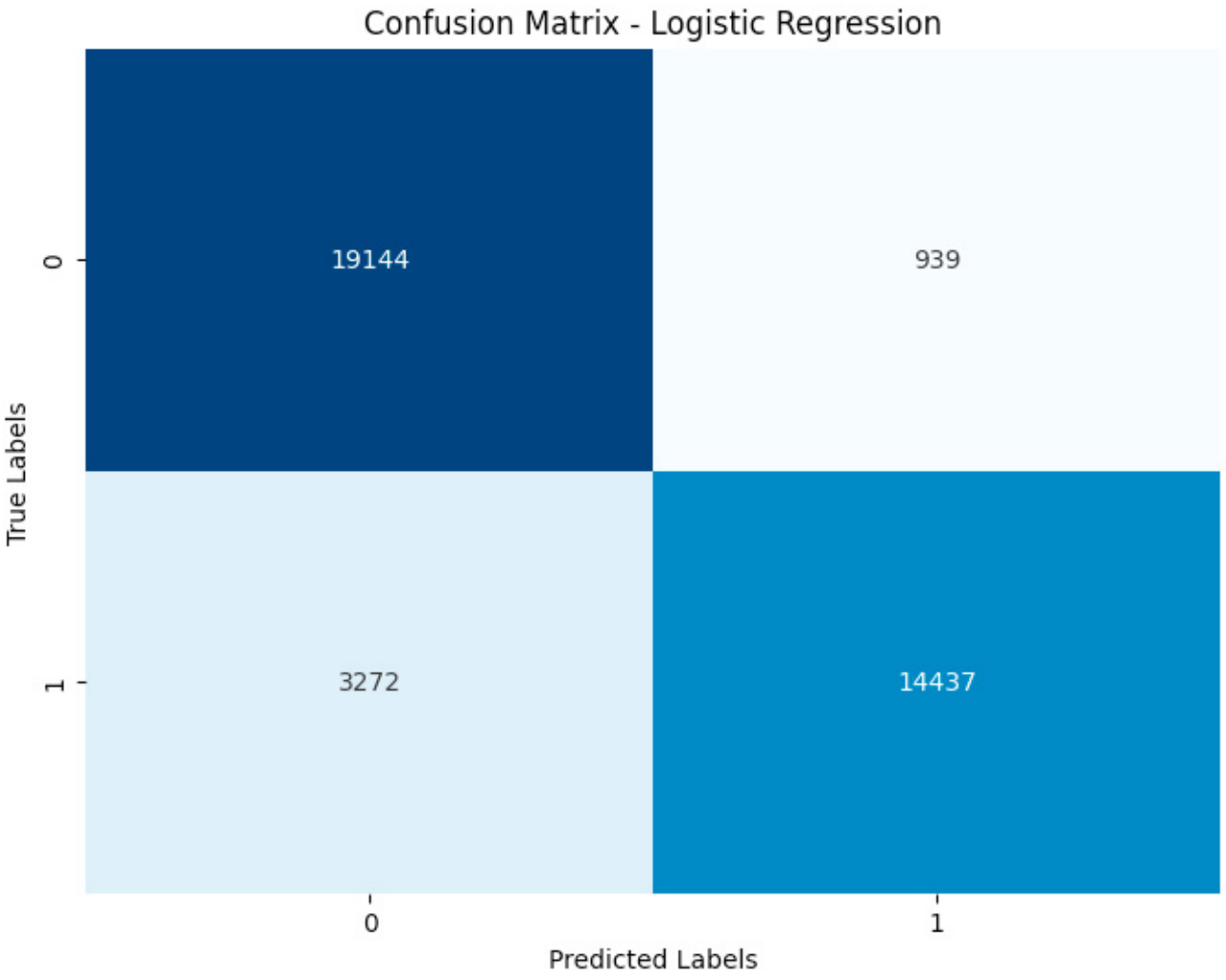


Fig. 2. The confusion matrix generated using the logistic regression executed on the validation dataset (where 'normal': 0, 'anomaly': 1).

Table 1. The accuracy and misclassification metrics for the proposed models.

Models	SVM (%)	Logistic regression (%)	KNN (%)
Accuracy	87.58	88.86	98.24
Misclassification	12.42	11.14	1.76

KDD dataset. It is clear from Table 2 that the proposed models have enhanced the accuracy and performance by choosing the essential features.

The precision rate of the SVM, logistic regression, and KNN models were 86.91, 87.83, and 97.99%, respectively. The recall was similar for SVM (87.31%) and logistic regression (87.98%) models. In addition, the f1-score was 87.07 and 88.48% for SVM and logistic regression, respectively. It was noticed that the highest precision rate (97.99%), recall (97.91%), and f1-score (98%) were obtained using the KNN model, which is higher than SVM and logistic regression. The confusion matrices have been generated on the validation datasets to show the effectiveness and robustness of the proposed approaches. Figures 1, 2, and 3 demonstrate the confusion matrix generated using the

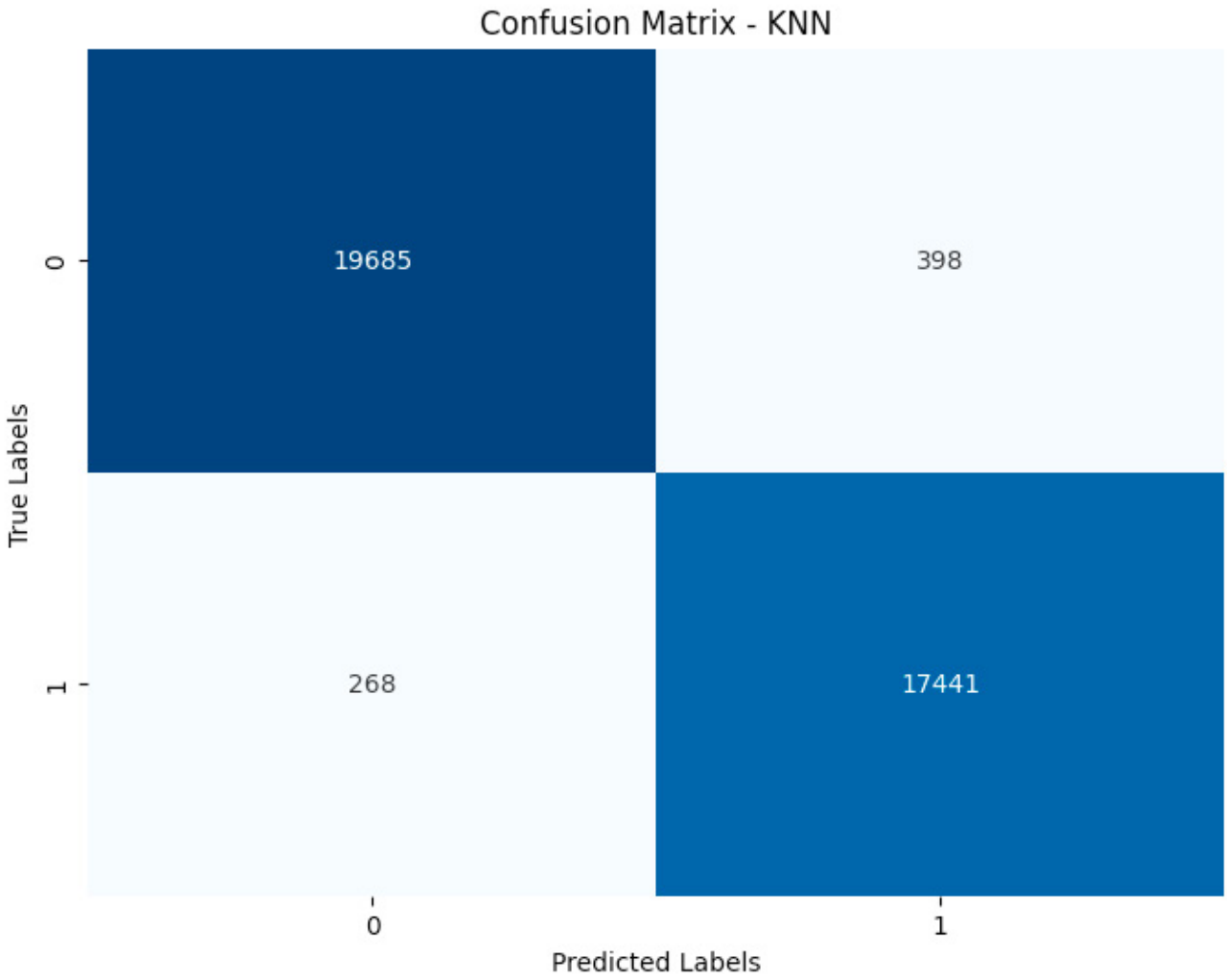


Fig. 3. The confusion matrix generated using the KNN method executed on the validation dataset (where, 'normal': 0, 'anomaly': 1).

Table 2. The evaluation results of the NSL-KDD dataset with selected features for binary classification.

Classifiers	Accuracy	Precision	Recall	F1-Score
SVM	87.58	86.91	87.31	87.07
Logistic regression	88.86	87.83	87.98	88.48
KNN	98.24	97.99	97.91	98.00

SVM, logistic regression, and KNN methods. In the case of SVM, there was a misclassification between the regular class (0) and the intrusion class (1). A total of 4,695 (12.42%) values were misclassified with each other, which was the reason for moderate accuracy in SVM classification (see Figure 1). However, the logistic regression has also produced similar misclassification between the regular and intrusion classes with 4,211 (11.14%) values (Figure 2). In the case of KNN, negligible misclassification was found to be only 1.76% (666 values). However, there are more correctly classified instances in all the methods.

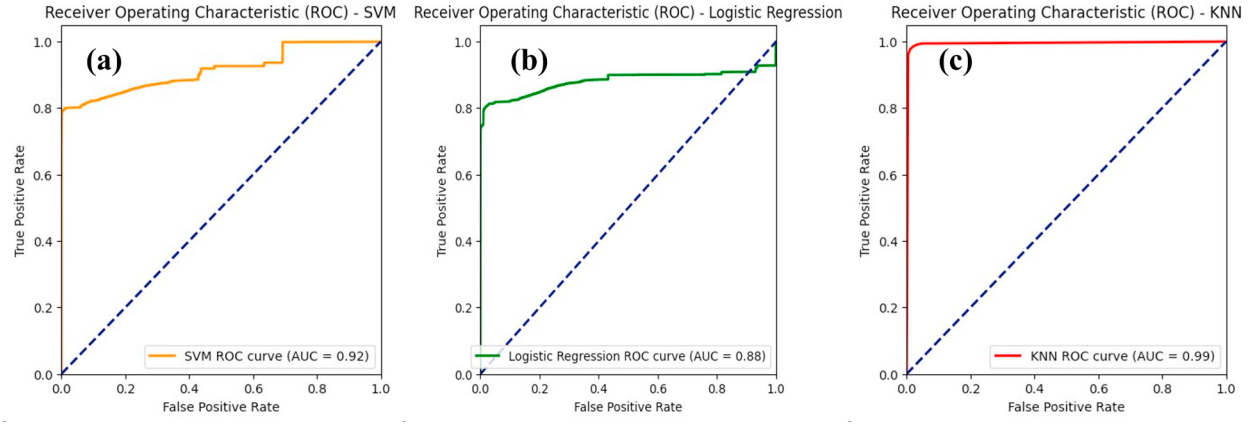


Fig. 4. The (a) SVM, (b) logistic regression, and (c) KNN models receiver operating characteristics curve.

The validity of the results was also confirmed with the AUC-ROC curves shown in Figures 4-(a), (b), and (c) for SVM, logistic regression, and the KNN methods. The ROC curve is an evaluation measurement to compute the classification accuracies that can be calculated via real and incorrect optimistic rates at particular threshold values [18]. We used a 10-fold cross-validation approach in our proposed models to compute and plot the AUC-ROC curves (Figure 4). The areas under the ROC curves were 0.92, 0.88, and 0.99 for SVM, logistic regression, and the KNN models (Figures 4-(a), (b), and (c), respectively). It is seen from the AUC-ROC curves that the proposed models correctly resulted in each class with the highest recognition rate. The KNN's ROC curve has better classification accuracy and logistic regression than SVM.

4.1. Comparative study with the existing literature

The significance of the presented strategy is compared with the existing standard literature. It is observed that the feature selection methods are essential in selecting suitable and accurate features. Some studies have focused on traditional feature selection methods and achieved satisfactory results. In a study [8], correlation-based feature selection earned 90.38 and 91.33% accuracy through SVM and KNN, less than in the present study. The accuracy of the KNN model in the study [1] is 95.39% when using the fuzzy-based feature selection approach. The proposed method of the current studies is better than that of earlier [1, 2, 4, 5, 8, 11, 12] studies regarding accuracy, training, and testing timings as per our results. Table 3 depicts the importance of the present study compared with recent literature.

Table 3. The comparison of the present study with recent literature.

Datasets	Feature Extraction/Selection	Machine/Deep Learning Models	Accuracy (in %)	Reference
NSL-KDD	Fuzzy-based approach	KNN and SVM	95.39 and 96.89	[1]
NSL-KDD	UEFFS	Ensemble learning	96.06	[2]
NSL-KDD	XGBoost	Deep learning	88.13	[4]
NSL-KDD	NA	C5 and one-class SVM	83.24	[5]
NSL-KDD	Correlation	SVM and KNN	90.38 and 91.33	[8]
NSL-KDD	logistic regression	2-class SVM and bagging	69.52 and 97.51	[11]
KDD CUP 99	NA	KNN and BiLSTM	78.1 and 95.4	[12]
NSL-KDD	**Random forest**	**SVM, logistic regression, and KNN**	**87.58, 88.86, and 98.24**	**This study**

5. Conclusions and future scope

The significant contribution of the present study is developing the network anomaly detection system using machine learning modelling. The proposed random forest algorithm has reduced the feature sizes from forty-one to nine with 99.54% accuracy. Subsequently, we used the nine features in training the proposed SVM, logistic regression, and KNN models and obtained 87.58, 88.86, and 98.24% testing accuracies. It is concluded that the proposed ensemble learning-based random forest feature selection approach is promising in selecting the best features. In addition, proposed machine learning models have provided the best detection rate with selected features compared to traditional methods. Our proposed KNN model has produced satisfactory performance in detecting and classifying network anomalies with a 98.24% success rate. In the future, the present study can be extended with deep learning-based models and more complicated real-time datasets to detect network intrusions that possess accuracy, efficiency, scalability, adaptability, and compliance to meet the challenges of evolving cyber-attacks while enabling ongoing advancements in the field.

Acknowledgements

The authors would like to acknowledge the data provider to conduct this research.

References

[1] Shiravani, A., Sadreddini, M. H., Nahook, H. N. (2023). "Network intrusion detection using data dimensions reduction techniques." *Journal of Big Data*, **10** (1), 27.
[2] Krishnaveni, S., Sivamohan, S., Sridhar, S. S., and Prabakaran, S. (2021). "Efficient feature selection and classification through ensemble method for network intrusion detection on cloud computing." *Cluster Computing* **24** (3): 1761-1779.
[3] Jaber, A. N., Rehman, S. U. (2020). "FCM–SVM based intrusion detection system for cloud computing environment." *Cluster Computing*, **23**, 3221-3231.
[4] Kasongo, S. M. (2023). "A deep learning technique for intrusion detection system using a Recurrent Neural Networks based framework" *Computer Communications*, **199**, 113-125.
[5] Khraisat, A., Gondal, I., Vamplew, P., Kamruzzaman, J., Alazab, A. (2020). "Hybrid intrusion detection system based on the stacking ensemble of c5 decision tree classifier and one class support vector machine." *Electronics*, **9** (1), 173.
[6] Tavallaee, M., Bagheri, E., Lu, W., Ghorbani, A. A. (2009, July). "A detailed analysis of the KDD CUP 99 data set." *In 2009 IEEE symposium on computational intelligence for security and defense applications* (pp. 1-6). IEEE.
[7] Serinelli, B. M., Collen, A., Nijdam, N. A. (2020). "Training guidance with kdd cup 1999 and nsl-kdd data sets of anidinr: Anomaly-based network intrusion detection system." *Procedia Computer Science*, **175**, 560-565.
[8] Mari, A. G., Zinca, D., Dobrota, V. (2023). "Development of a Machine-Learning Intrusion Detection System and Testing of Its Performance Using a Generative Adversarial Network." *Sensors*, **23** (3), 1315.
[9] Halimaa, A., Sundarakantham, K. (2019, April). "Machine learning based intrusion detection system." *In 2019 3rd International conference on trends in electronics and informatics (ICOEI)* (pp. 916-920). IEEE.
[10] Aljawarneh, S., Yassein, M. B., Aljundi, M. (2019). "An enhanced J48 classification algorithm for the anomaly intrusion detection systems." *Cluster Computing*, **22**, 10549-10565.
[11] Besharati, E., Naderan, M., Namjoo, E. (2019). "LR-HIDS: logistic regression host-based intrusion detection system for cloud environments." *Journal of Ambient Intelligence and Humanized Computing*, **10**, 3669-3692.
[12] Gao, J. (2022). "Network intrusion detection method combining CNN and biLSTM in cloud computing environment." *Computational Intelligence and Neuroscience*, 2022.
[13] https://www.unb.ca/cic/datasets/nsl.html Accessed on 14 August 2023.
[14] Kumar, G. S. C., Kumar, R. K., Kumar, K. P. V., Sai, N. R., Brahmaiah, M. (2023). "Deep Residual Convolutional Neural Network: An Efficient Technique for Intrusion detection system." *Expert Systems with Applications*, 121912.
[15] Alotaibi, S. D., Yadav, K., Aledaily, A. N., Alkwai, L. M., Yousef Dafhalla, A. K., Almansour, S., Lingamuthu, V. (2022). "Deep neural network-based intrusion detection system through PCA." *Mathematical Problems in Engineering*, 2022.
[16] Li, X., Chen, W., Zhang, Q., Wu, L. (2020). "Building auto-encoder intrusion detection system based on random forest feature selection." *Computers Security*, **95**, 101851.
[17] Kasongo, S. M., Sun, Y. (2020). "Performance analysis of intrusion detection systems using a feature selection method on the UNSW-NB15 dataset." *Journal of Big Data*, **7**, 1-20.
[18] Luo, J., Chai, S., Zhang, B., Xia, Y., Gao, J., Zeng, G. (2020). "A novel intrusion detection method based on threshold modification using receiver operating characteristic curve." *Concurrency and Computation: Practice and Experience*, **32**(14), e5690.